

Affine Richardson Geometry: Clusters, Links, and Positroid Patches

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1 Introduction

In finite type, cluster structures and Richardson links have turned questions about coordinate rings, point counts, and cohomology into problems in diagrams and knot theory [GL23, GL24, GLSBS, CGGLSS]. I propose to extend these connections to affine type A , providing comparable tools for individual positroid patches and explaining how their geometry changes with the choice of Grassmannian chart.

The Grassmannian $\mathrm{Gr}(k, n)$ is covered by affine charts $\{\Delta_I \neq 0\}$, indexed by k -subsets I , and stratified by open positroid varieties Π_f° . Their intersections are the *positroid patches* $U_{f,I} := \Pi_f^\circ \cap \{\Delta_I \neq 0\} \cong \mathring{R}_{f,t_I}$, identified by Snider with open Richardson varieties in the affine flag variety; f is a bounded affine permutation and t_I the translation determined by the chart [Sni10]. Two patches of the same positroid variety can have different geometry. For the top open positroid variety in $\mathrm{Gr}(2, 4)$, my joint work with Matthew J. Tyler gives

$$U_{f,13} \cong (\mathbb{C}^*)^4, \quad \#U_{f,13}(\mathbb{F}_q) = (q-1)^4, \quad \#U_{f,12}(\mathbb{F}_q) = (q-1)^4 + q(q-1)^2.$$

The first patch is a torus and the second is not [MT26]. The global positroid link records the point count of the underlying variety [GL24] and cannot distinguish these patches. An affine theory must retain I .

The setting is the family of *affine braid Richardson varieties* $X_{f,\beta}$, indexed by a positive affine braid word β and $f \leq \delta(\beta)$, where δ is the Demazure product. They parametrize sequences of affine flags with prescribed simple relative positions. Reduced β representing g gives $\mathring{R}_{f,g}$, including $U_{f,I}$ when $g = t_I$. For extended affine indices, I remove the common length-zero factor before taking braid lifts. My objectives address their coordinate rings, topology through link invariants, and the diagrammatic meaning of weak separation. Together they would explain when ordinary positroid constructions suffice and provide tools for the remaining affine patches.

2 Past accomplishments

I introduced *affine Deodhar diagrams*, or affine Deograms, recording a bounded affine permutation and a lattice path specifying the chart [Mar26]. They encode the Billig–Dyer decomposition into strata $(\mathbb{C}^*)^{e(D)} \times \mathbb{C}^{b(D)}$ [BD94]. Distinguished subexpressions give the point count

$$R_{f,\beta}(q) := \#X_{f,\beta}(\mathbb{F}_q) = \sum_D (q-1)^{e(D)} q^{b(D)}.$$

My local moves explain point-count identities, and my recursive bijections connect maximal Deograms with rational Dyck paths. These results provide a starting point for seeking geometric interpretations of affine Catalan combinatorics, beyond its enumeration.

With Tyler, I proved an affine extension of the finite-type torus criterion of Gorsky–Kim–Sherman-Bennett [GKSB26], and a topological criterion for the standard torus action.

Theorem 1 (Martinez–Tyler [MT26, Theorem 1.1]). *Let $f \leq g$ be affine permutations and $d = \ell(g) - \ell(f)$. The following are equivalent:*

- (i) $\mathring{R}_{f,g} \cong (\mathbb{C}^*)^d$;
- (ii) $[f, g]$ contains no 2-crown;
- (iii) $\mathring{R}_{f,g}$ has a single Deodhar stratum.

Theorem 2 (Martinez–Tyler [MT26, Theorem 1.3]). *For a finite positive braid word β and $u \leq \delta(\beta)$, $X_{u,\beta}$ has a dense standard-torus orbit if and only if its Richardson link $L_{u,\beta}$ is smoothly slice.*

These results distinguish two notions of toricity: a variety may be isomorphic to an algebraic torus even when its standard torus action has no dense orbit. The torus cases have Laurent polynomial coordinate rings, making them natural starting examples for the proposed affine cluster structure. I will seek to realize these rings using *fully frozen* seeds, with no mutable variables.

With Moriah Elkin and Tyler, I construct candidate quivers $Q_{f,\beta}^{\text{cand}}$ from affine 3D plabic diagrams and prove a bijection between affine Le-diagrams and equivalence classes of reduced punctured plabic graphs [EMT26]. We also establish the following.

Theorem 3 (Elkin–Martinez–Tyler [EMT26]). *For every positive affine braid word β and $f \leq \delta(\beta)$, the candidate quiver $Q_{f,\beta}^{\text{cand}}$ is locally acyclic.*

The geometric identification of this quiver with a seed for $X_{f,\beta}$ remains conjectural. Our affine Le-diagram characterization and weak-separation criterion appear in Objective 3.

3 Research objectives, methods, and significance

Objective 1: Affine cluster structures

Finite-type positroid and braid varieties admit cluster structures in which combinatorial moves describe changes of coordinates [GL23, GLSBS, CGGLSS]. An affine extension would express every regular function on $X_{f,\beta}$ in terms of cluster variables and inverses of frozen variables, with diagrams determining their exchange relations. This would give explicit coordinate models for positroid patches, including those beyond ordinary plabic face constructions.

Bao–Ye construct a seed $\Sigma_{f,\beta}^{\text{BY}}$ with $\mathbb{C}[X_{f,\beta}] = \mathcal{U}(\Sigma_{f,\beta}^{\text{BY}})$, frozen variables inverted [BY25, Theorem 6.5]. An upper cluster algebra records functions Laurent in every cluster; the missing step is to show that cluster variables generate this ring. Our locally acyclic candidate quiver supplies a route to that equality, once it is identified with a geometric seed [Mul13].

Conjecture 4 (Elkin–Martinez–Tyler [EMT26]). *Let β be a positive affine braid word and $f \leq \delta(\beta)$. There is a seed $\Sigma_{f,\beta}$ in $\mathbb{C}(X_{f,\beta})$ with ice quiver $Q_{f,\beta}^{\text{cand}}$ such that, with frozen variables inverted,*

$$\mathbb{C}[X_{f,\beta}] = \mathcal{A}(\Sigma_{f,\beta}) = \mathcal{U}(\Sigma_{f,\beta}).$$

I will compare our diagrammatic construction with Bao–Ye’s seed, matching generalized-minor functions, exchange matrices, and frozen variables in a common function field. Torus cases and quivers with one mutable vertex provide initial tests before translation endpoints and general affine words. If the seeds differ, I will seek mutations or a quasi-cluster comparison with compatible frozen localizations, verifying the resulting identification of coordinate rings.

Objective 2: Affine Richardson links

Richardson links give access to geometric information beyond point counts. Galashin–Lam identify the Hochschild-degree-zero (top a -graded) part of Khovanov–Rozansky homology with torus-equivariant compactly supported cohomology of open Richardson varieties, after regrading [GL24, Theorem 1.17]. Using torus-knot computations, they determine the mixed Hodge polynomial of the top open positroid variety $\Pi_{k,n}^\circ$: for $\gcd(k, n) = 1$,

$$\mathcal{P}(\Pi_{k,n}^\circ; q, t) = (q^{1/2} + t^{1/2})^{n-1} C_{k, n-k}(q, t),$$

where $C_{k, n-k}(q, t)$ is the rational q, t -Catalan polynomial and $\mathcal{P}$ uses their mixed Hodge grading [GL24, Theorem 1.1]. This computes Betti numbers and refines point counts. Combined with curious Lefschetz from cluster geometry, it yields Catalan symmetry and unimodality [GL24, Corollary 1.8].

An affine counterpart could provide computations of weight-graded cohomology for individual positroid patches and geometric interpretations of refinements of my Deodhar and Dyck-path combinatorics. The first step is a link invariant recovering the point count. For $v \leq w$ in S_n , $L_{v,w}$ is the closure of $\beta(w)\beta(v)^{-1}$, and a normalized top HOMFLY coefficient gives the point count [GL24, Theorem 2.1]. I use $P_\circ = 1$ and $aP_+ - a^{-1}P_- = zP_0$. My candidate $L_{f,\beta}$ has full periodic lift $\beta\beta(f)^{-1}$ to the universal cover of its ambient space. It specializes to a candidate $L_{f,I}$ for each patch, retaining the chart data that distinguishes the two $\text{Gr}(2, 4)$ examples.

Conjecture 5 (Martinez). *For the candidate $L_{f,\beta}$, there is a uniform HOMFLY skein evaluation and normalization of the closure extending the finite-type convention such that its normalized top a -coefficient satisfies*

$$\mathcal{H}_{f,\beta}^{\text{top}}(q) = R_{f,\beta}(q) = \#X_{f,\beta}(\mathbb{F}_q).$$

The evaluation and normalization depend only on the braid and closure data.

I have checked the finite-type specializations of Conjecture 5 for all reduced braid lifts of permutations in S_6 , using SageMath [Sage]. I will first make the periodic closure and its HOMFLY evaluation precise and establish diagram invariance, using affine Hecke traces and monotone torus links as models [GN22, GLMon23]. Comparing skein relations with distinguished-subexpression recursions should prove the conjecture for families closed under these reductions.

For the homological extension, I will begin with tori and short affine words, comparing link homology and equivariant cohomology with the required grading shifts and torus factors. A successful correspondence should recover Conjecture 5 by graded Euler characteristic and suggest diagram statistics encoding cohomological degree and Hodge weight. I will also investigate which affine patches satisfy the hypotheses of cluster-theoretic Lefschetz results and whether these yield symmetries of the resulting polynomials, as it does in the finite case.

Objective 3: Affine positroid combinatorics

Ordinary Le-diagrams index positroid cells in the totally nonnegative Grassmannian. Their dots record cell dimensions and positive parameters; their networks compute Plücker coordinates from weighted paths [Pos06].

For a fixed positroid, weak separation characterizes the indices that occur as ordinary plabic face labels [OPS15, Theorem 1.5]. Freezing these faces describes some positroid patches. My affine

models aim to provide coordinates for the remaining patches and explain the geometric role of weak separation.

Theorem 6 (Elkin–Martinez–Tyler [EMT26]). *Let $\mathcal{R}$ be the region associated with a periodic $k \times (n - k)$ path. A periodic dot filling of $\mathcal{R}$ is an affine Le-diagram if and only if every southwest–northeast pair of dots whose square is contained in $\mathcal{R}$ has a dot at its southeast corner.*

The additional Le-constraint forbids any southwest–northeast pair of dots whose square is not contained in $\mathcal{R}$. The associated “soap films” organize the candidate seed functions.

Theorem 7 (Elkin–Martinez–Tyler [EMT26]). *Let $U_{f,I}$ be a nonempty positroid patch, $D_{f,I}$ its associated affine Deogram, and $\mathcal{I}_f$ the Grassmann necklace. The following are equivalent:*

- (i) $D_{f,I}$ satisfies the additional Le-constraint;
- (ii) I is weakly separated from every $J \in \mathcal{I}_f$;
- (iii) the soap films of $D_{f,I}$ do not intersect.

Theorem 7 explains weak separation through periodic diagrams and soap-film geometry. Its proposed geometric consequence is a Plücker description of the candidate seed.

Conjecture 8 (Elkin–Martinez–Tyler [EMT26]). *Suppose $D_{f,I}$ satisfies the additional Le-constraint, and fix $J_0 \in \mathcal{I}_f$. For each variable x_v of its candidate affine seed, there are $J_v \in \binom{[n]}{k}$ and an invertible frozen monomial m_v such that*

$$x_v = m_v \frac{\Delta_{J_v}}{\Delta_{J_0}}.$$

I will compare this seed with the one obtained by freezing the face labeled I , matching functions and exchange data and realizing $\mathbb{C}[U_{f,I}] = \mathbb{C}[\Pi_f^\circ][(\Delta_I/\Delta_{J_0})^{-1}]$. Moves preserving the chosen face give the finite-type model [OS17].

When I is not weakly separated from the necklace, no ordinary plabic graph for this positroid has face label I , although Δ_I remains a cluster monomial [MMSV24, Theorem 8.9]. Beginning with a single violation of the additional Le-constraint, I will compute generalized minors and exchange relations from intersecting films, and express these functions in the localization obtained by inverting the factors of Δ_I . These formulas would extend the ordinary plabic coordinate model to patches unavailable by freezing a single face.

[figure here?](#)

Work plan and assessment. My initial plan is to spend the first year comparing the proposed affine seeds with Bao–Ye’s construction in basic examples and making the affine link construction precise. The second year will focus on coordinates for positroid patches and the point-count correspondence for selected families; the third on link homology, cohomology, and affine Deodhar combinatorics. I will regularly reassess these priorities with Sherman-Bennett, using new examples, results, and developments in the field to refine the questions and adjust the order of work.

4 Career development

I will take the Research Instructorship option to develop an independent research program alongside my teaching. Regular meetings with Sherman-Bennett on Objective 1 will build expertise in generalized minors, cluster constructions, and coordinate rings, enabling me to turn my affine combinatorial models into geometric theories.

Objective 2 will extend my training to skein methods and Khovanov–Rozansky homology through the Davis topology community. I plan to organize a learning seminar on the finite-type results and present work in the Algebra and Discrete Mathematics seminar. Talks and manuscripts for audiences in several fields will strengthen my communication and develop new collaborations. Teaching will build my skills in course design, exposition, and assessment; undergraduate mentoring will develop the supervision and project-design skills needed for an academic career.

5 Sponsoring scientist and host institution

UC Davis brings together the expertise needed for all three objectives. Melissa Sherman-Bennett co-developed 3D plabic graphs and the Deodhar approach to braid-variety cluster structures [GLSBS]. Her comparison of cluster constructions [Comp25] provides the finite-type model for Objective 1; her work with Gorsky and Kim on Richardson tori [GKSB26] is the criterion extended by Theorem 1. These connections make her a natural sponsor for the geometric development of my diagrams.

Eugene Gorsky’s work on braid varieties, link homology, and affine Hecke traces supports Objective 2 [CGGLSS, GN22]. Roger Casals’s work on braid-variety cluster structures and Legendrian weaves supplies a contact and symplectic perspective [CGGLSS, Comp25]. Their approaches will help me interpret the affine diagrams geometrically and develop the link construction.

6 Broader Impacts

Undergraduate research shaped my decision to pursue graduate study. Since 2023, I have mentored eight undergraduates through the UCLA Directed Reading Program and served on its organizing committee. In spring 2026, Olha Shevchenko and I co-mentored three of these students on a project that led to a joint preprint on unimodality of q -Fibonomial coefficients [CIMS26].

At Davis, I aim to mentor one or two undergraduates annually through the Davis Math Lab and directed reading. Projects on affine Deodhar diagrams and rational Dyck paths will offer accessible problems in enumeration and bijective combinatorics. I will tailor the work to students’ preparation, guiding them from examples to conjectures and proofs to teach them about the process of mathematical research.

I will also begin mentoring graduate students through joint reading and collaborative research, offering guidance on approaching the literature, choosing tractable questions, and preparing talks and manuscripts. I will share advice on navigating graduate study and work with graduate mentors on designing and supervising undergraduate projects.